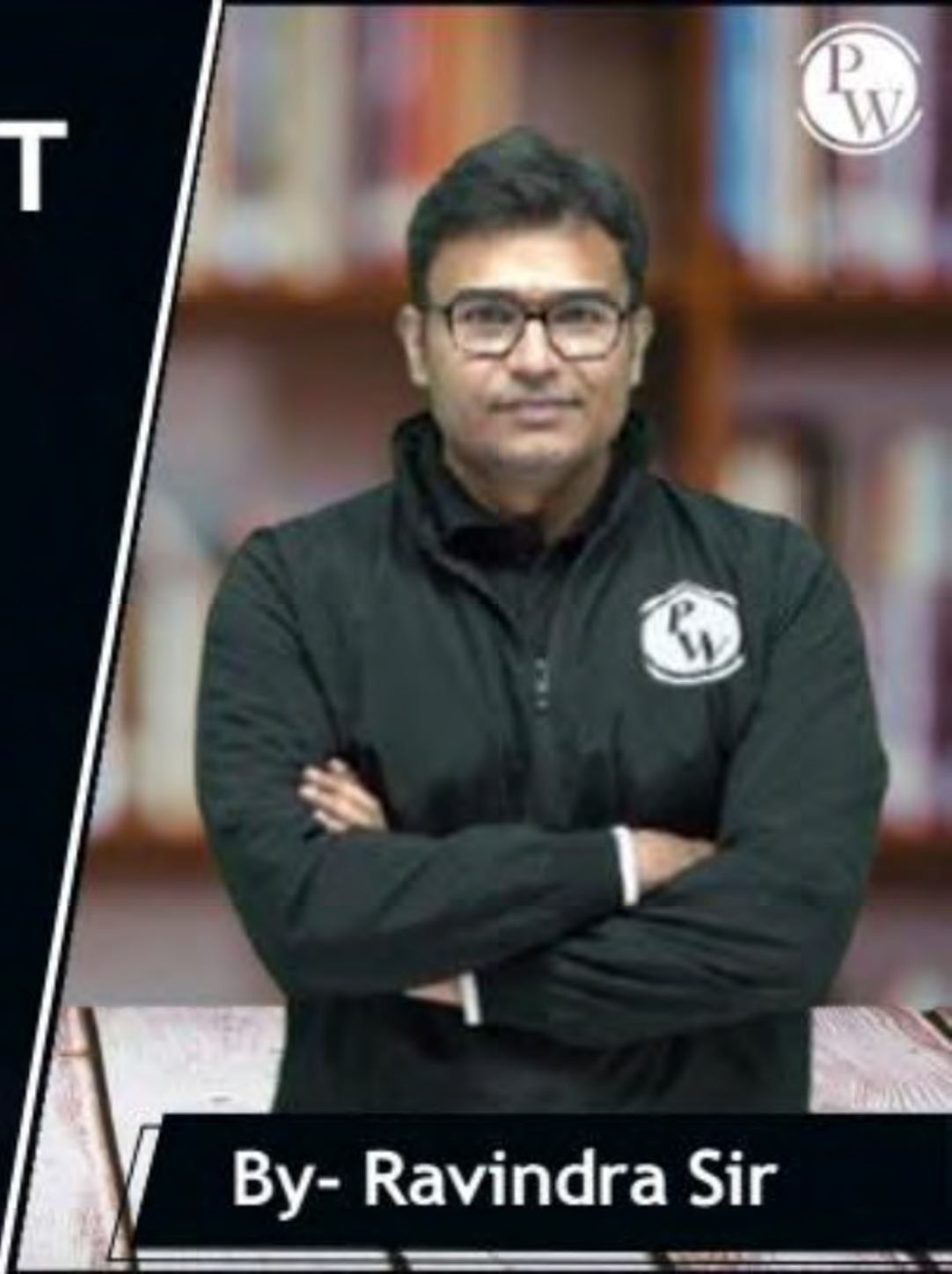
Computer Science & IT

COMPUTER NETWORK (CN)

Application Layer Protocols

Lecture No. 03



By- Ravindra Sir

Recap of Previous Lecture



- Topic HTTP
- Topic
- Topic

Topics to be Covered



- Topic — HTTP
- Topic — NAT ✓
- Topic

Proxy Server location :-

The proxy servers are normally located at the client side

1) The client computer can also be used as a proxy server, in small capacity, that stores responses to requests often invoked by the client.

2) In a company, a proxy server may be installed on the N/w LAN to reduce the load going out and coming into the LAN.



3) An ISP with many customers can install a proxy server to reduce load going out and Coming into ISP N/W

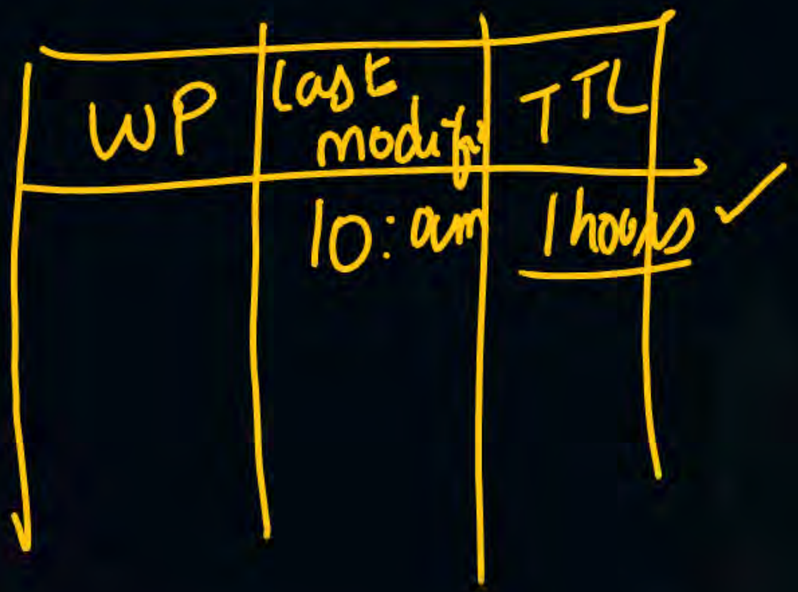
Cache update :-

How long a response should remain in the proxy server before being deleted and replaced.

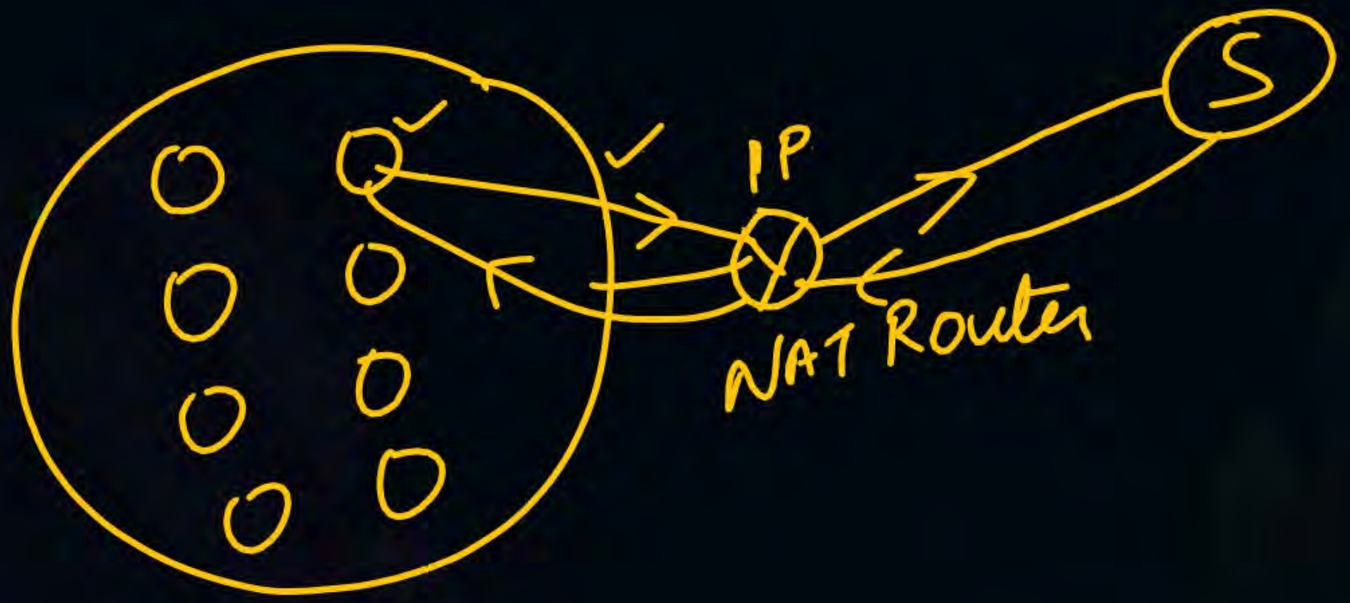
One solution is to store list of sites whose information remains same for a while.

Ex :- A news agency may change its news papes every morning. This means that a proxy server can get the news early in the morning and keep it until the next days.

Another Solution is to add some headers to show the last modification time of the information. The proxy server can then use the information in the header to guess how long the information would be valid



Network addres translation (NAT):-



Private addresses :-



10.0.0.0 to 10.255.255.255 ✓

172.16.0.0 to 172.31.255.255 ✓

192.168.0.0 to 192.168.255.255 ✓

The distribution of addresses through ISP's have created a new problem. Assume that an ISP has granted a small large of IP add to a small business of a household. If the business grows of the household need a large Arange, the ISP may not be able

able to grant the demand because the addresses before and after the range may already have been allocated to other n/w

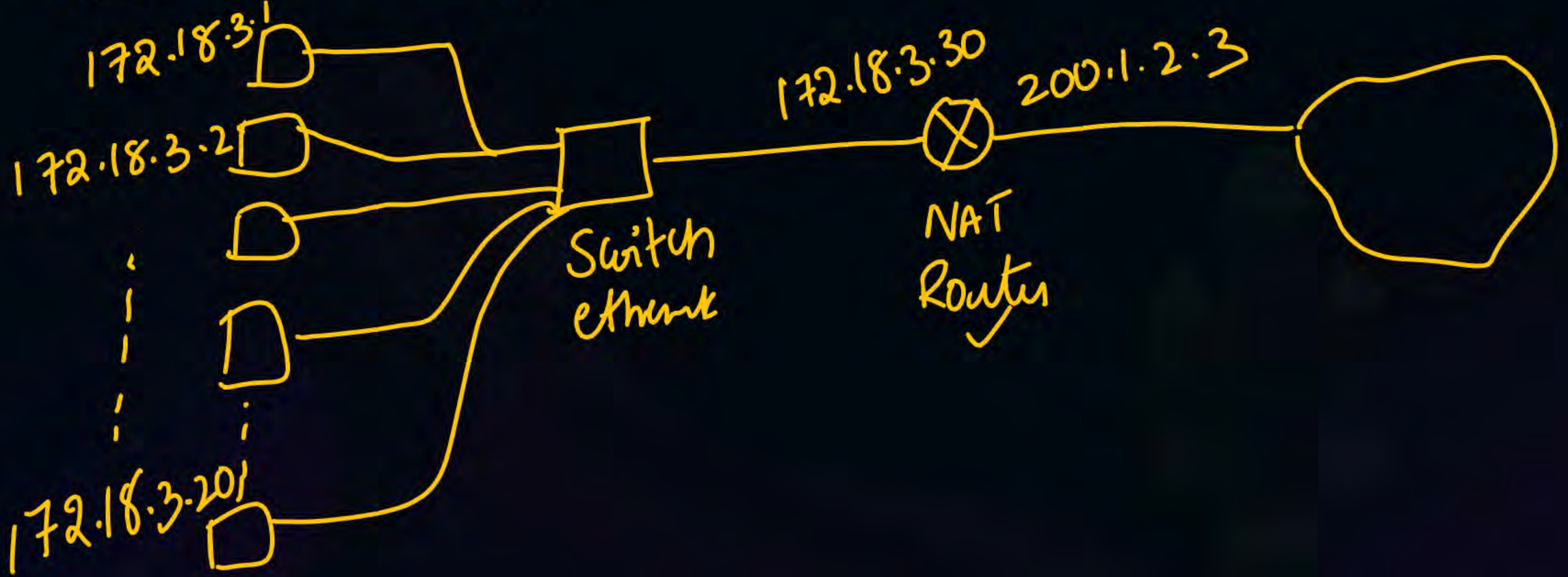


In most situations, however only a portion of computers in a N/w need to access the internet Simultaneous. This means that the number of allocated addresses does not have to match with the number of computers in the N/W. For example, assume in a small business with 20 computers, the maximum no of computers that access internet simultaneously are only 4. Most of the computers are either doing some task that does not need internet access or communicating with each other

This small company can use TCP/IP protocol for both internal and universal communication. The business can use 20 addresses from the private addresses for internal communication, 5 addresses for universal communication can be assigned by the ISP

→ A technology that can provide the mapping b/w private and public address, and at the same time support virtual private N/w (VPN) is Network address translation. This technology allows a site to use a set of private addresses for internal communication and a set of global Internet addresses for the communication to rest of the world. The site must have only one single connection to global Internet through a NAT capable router.

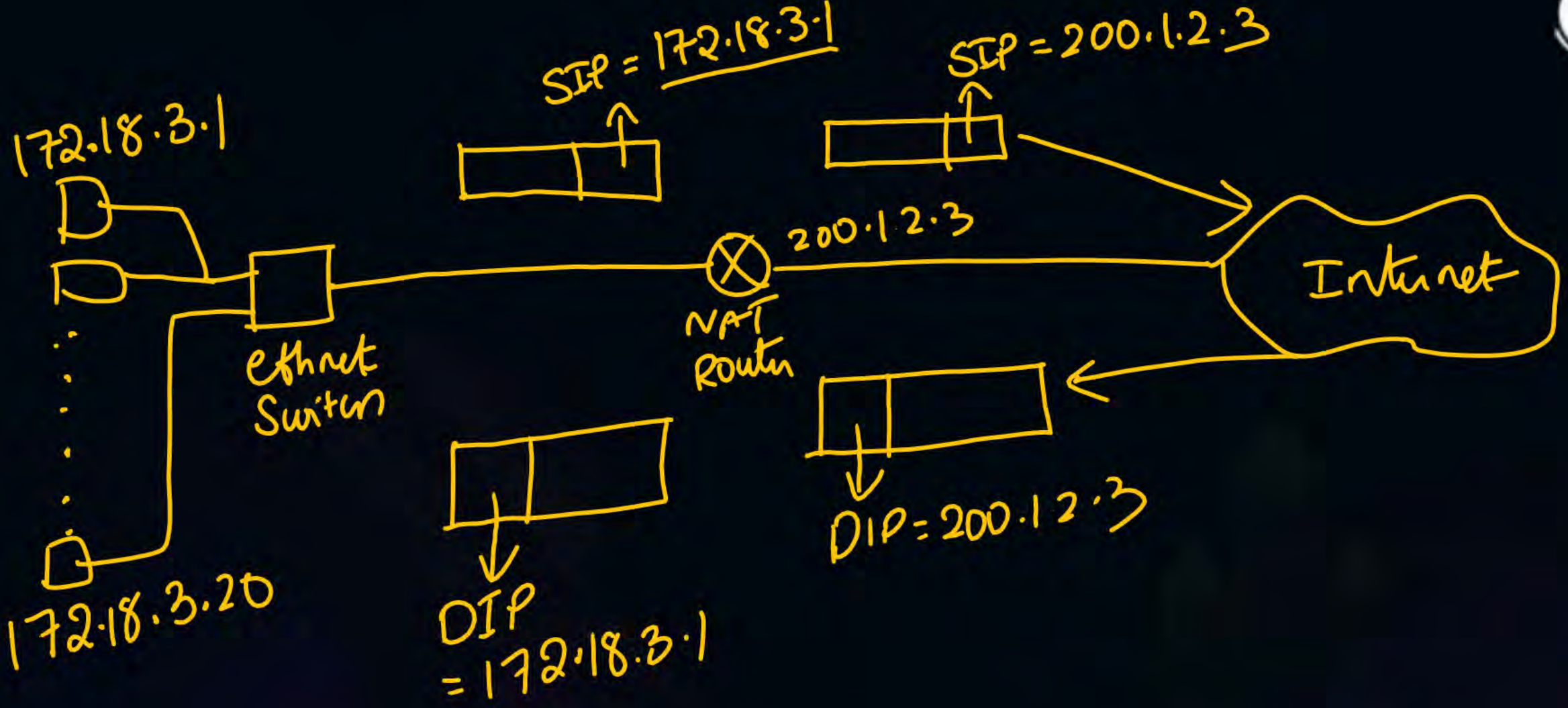
The private N/w uses private addresses. The router that connects the N/w to global addres uses one private addres and one global addres



The private N/W is not visible to the rest of the internet.
The rest of the internet sees only NAT router with
public address 200.1.2.3

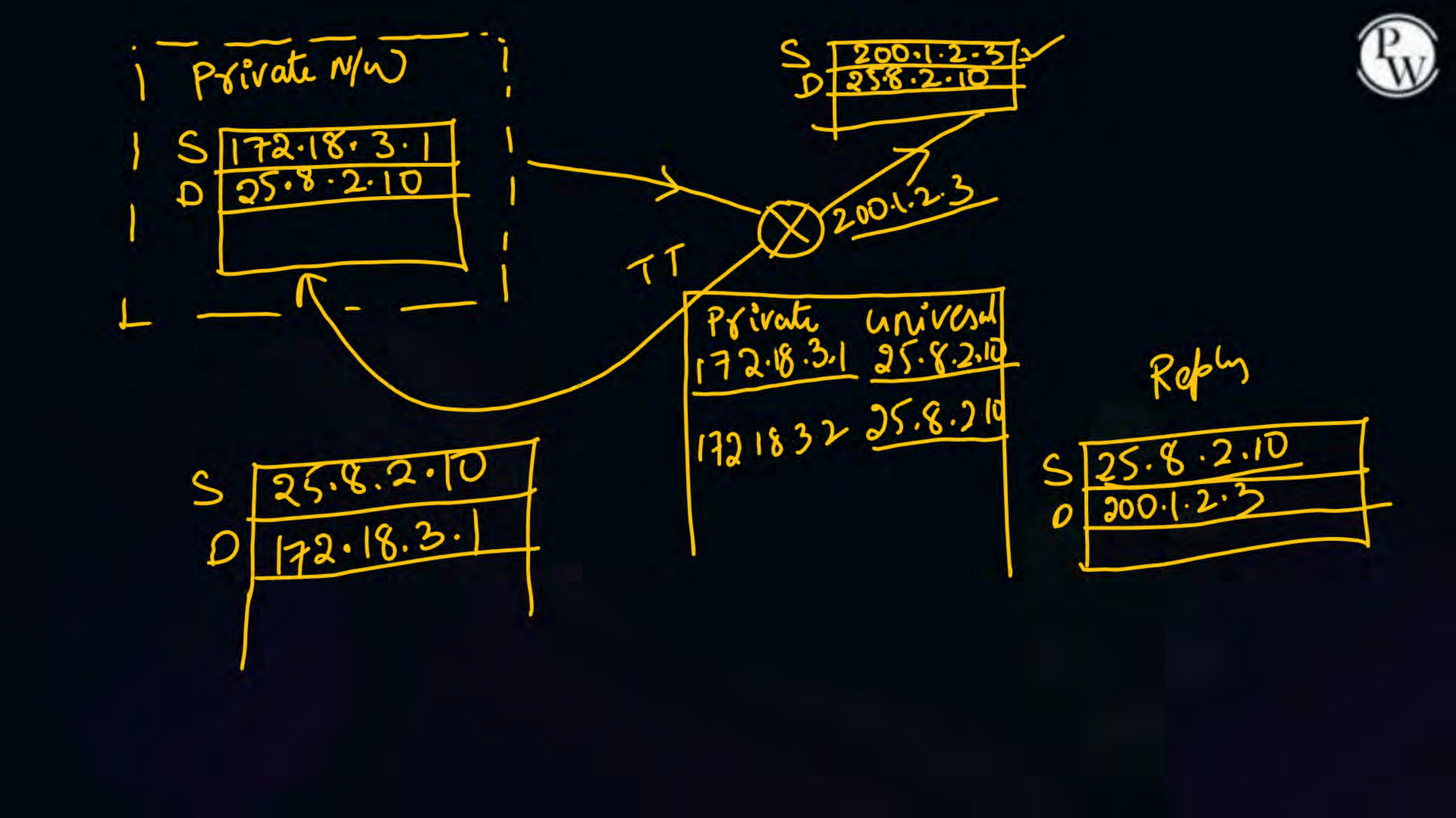
Addrs translation :-

All the outgoing packet go through NAT Router. which replaces Source address in the packet with global address. All incoming packets pass through NAT router, which replaces destination address in the packet with appropriate private address


SIP = 172.18.3.1
SIP = 200.1.2.3
172.18.3.1
200.1.2.3
Internet
ethernet Switch
NAT Router
DIP = 200.1.2.3
DIP = 172.18.3.1
172.18.3.20

Translating table :-

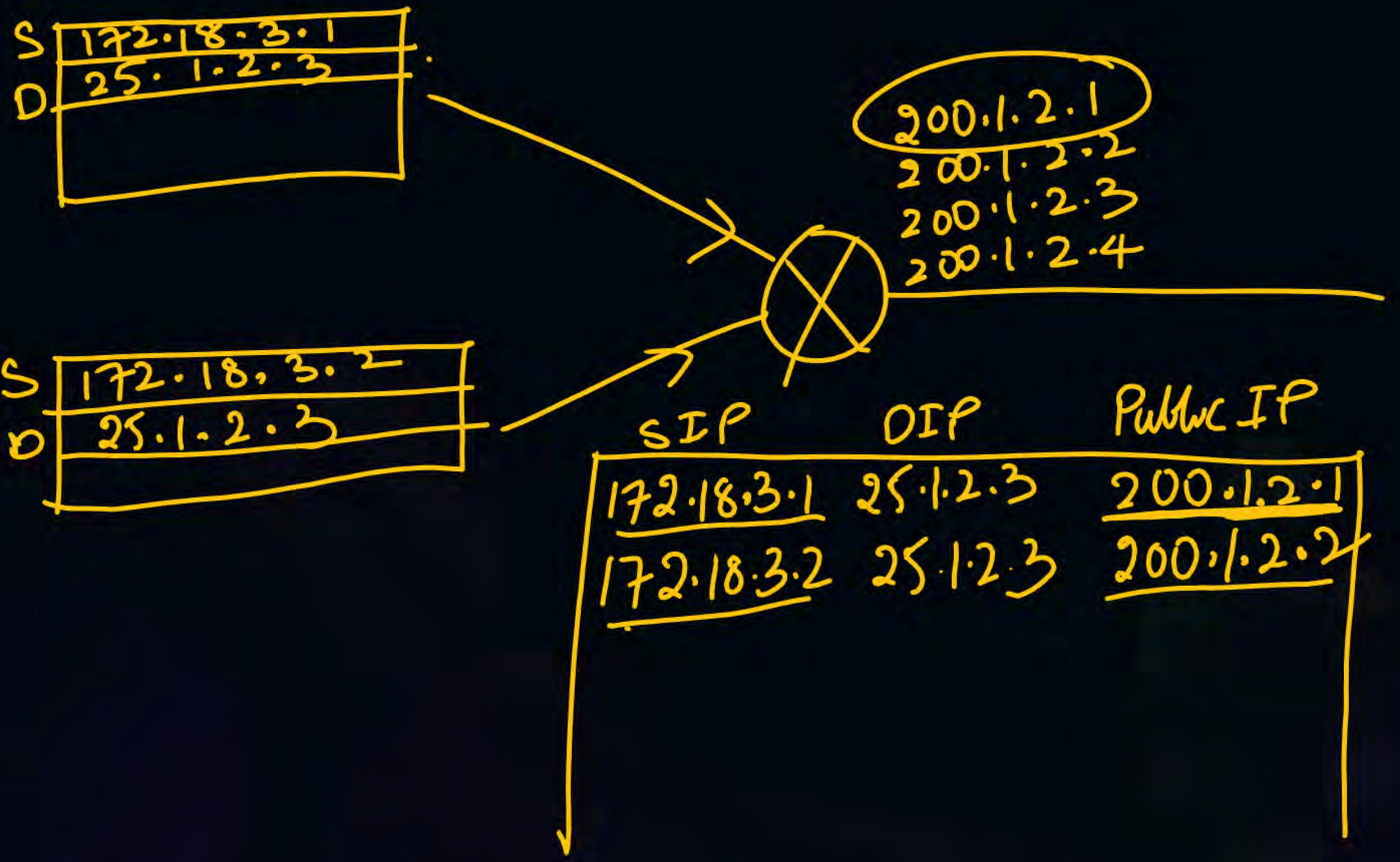
Translating the Source address to an outgoing packet is straightforward. But how does NAT router know the destination add of an incoming packet ? There may be tens or hundreds of private IP addresses each belonging to a specific host. This problem is solved if NAT router has a translation table


Private N/W
S 172.18.3.1
D 25.8.2.10
S 200.1.2.3
D 25.8.2.10
200.1.2.3
TT
Private Universal
172.18.3.1 25.8.2.10
172.18.3.2 25.8.2.10
S 25.8.2.10
D 172.18.3.1
Reply
S 25.8.2.10
D 200.1.2.3

In this strategy, Communication must always be initiated by the private N/W. NAT is mostly used by ISPs to assign private addresses to the Customers.
In this case communication with internet is always initiated from customer side, using a client program such as HTTP, TELNET & FTP. For example if email originates from outside the N/W, it is received by email server of ISP and stored in the mailbox of the customer until retrieved with a protocol such as POP by the private host

Using a pool of IP addrs :-

The use of only one global addrs by NAT Router allows only one private N/W host to access a given external host. To remove this restriction, NAT router can use a pool of global addresses. Instead of using one global addrs (200.1.2.3), NAT Router can use 4 addresses (200.1.2.1, 200.1.2.2, 200.1.2.3, 200.1.2.4). In this case four hosts at the same time can communicate with same external host


S 172.18.3.1
D 25.1.2.3
S 172.18.3.2
D 25.1.2.3
200.1.2.1
200.1.2.2
200.1.2.3
200.1.2.4
SIP OIP Public IP
172.18.3.1 25.1.2.3 200.1.2.1
172.18.3.2 25.1.2.3 200.1.2.2

In this case, only four hosts can communicate with same external host at the same time

Using both IP add and port addresses :-

To allow many to many relationship b/w private n/w host and external server programs, we need more information in translation table

Private add	Private PORT	extenal add	External port	Transport protocol
172.18.3.1	1400 ✓	25.8.3.2	80	TCP
172.18.3.2	1401 ✓	25.8 3.2	80	TCP

For this method to work, port number must be unique.

PYQ S

-> Traceroute reports a possible route that is taken by the packets moving from host A to some other host B. Which of the following options represent the technique used by traceroute to identify these hosts

a) By progressively querying routers about the next router on the path to host B using ICMP packets starting from first routers

b) Requiring each router to append the address to a ICMP packet as it is forwards to host B. The list of routers en-route to B is returned by B in an ICMP packet

c) By ensuring that an ICMP packet is returned to A by each router en route to B, in the ascending order of their hop distance from A.

d) By locally computing the shortest path from A to B

⇒ Count to infinity is a problem associated with

a) LSR
b) DVR
c) DNS
d) TCP

→ HELO and PORT, respectively are commands from the protocols

a) FTP and HTTP

b) Telnet and POP3

c) HTTP and Telnet

d) SMTP and FTP

Which of the following uses UDP?

a) HTTP b) Telnet c) DNS d) SMTP

THANK - YOU